SmartRoute Pro — ADA Algorithm Audit Report

Subject: Analysis and Design of Algorithms (ADA) • Dijkstra's Single-Source Shortest Path

1. Problem Statement & Mathematical Formulation

Let G = (V, E, w) be a directed, weighted graph representing the urban road delivery network, where V is the set of geographic hubs (vertices), E subset of V x V is the set of active road segments (edges), and w: E -> R+ assigns a strictly positive distance weight distance_km > 0 to each road segment. Given a designated source node s in V and a target destination t in V, find a directed simple path P = (v0, v1, ..., vk) such that v0 = s, vk = t, and the total travel distance sum(w(vi-1, vi)) is minimized.

2. Algorithmic Complexity & Min-Priority Queue Design

Data Structure: Adjacency list graph with Python's heapq (binary min-heap).
• **Theoretical Time Complexity:** O((|V| + |E|) log |V|) using a binary min-heap.
• **Theoretical Space Complexity:** O(|V| + |E|) for adjacency lists, tentative distance maps, and predecessor pointers.
• **Scope Limitation:** Dijkstra solves the single-pair shortest path problem optimally with non-negative edge weights. It does *not* solve the NP-hard Traveling Salesperson Problem (TSP) with multiple arbitrary permutations.

3. Empirical Route Run Benchmark

Benchmark Metric	Recorded Value
Route Run Identifier	**Run #2**
Algorithm Type	Dijkstra
Origin / Destination	**Zone 1: Tech Hub -> Zone 1: Tech Hub**
Optimal Shortest Distance	**0.00 km**
Algorithm Execution Time	**0.001 ms**
Optimal Node Traversal	[-> 2 ->]
Execution Timestamp	2026-10-16 15:41:22 UTC

4. Route Execution Audit Trail (Historical Log)

Run ID	Source Hub	Destination	Distance (km)	Runtime (ms)	Timestamp
#2	Zone 1: Tech Hub	Zone 1: Tech Hub	0.00	**0.00**	10-16 15:41
#1	Central Depot (Ori	Zone 5: Harbor Pie	15.90	**0.14**	10-16 15:41